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1 Adoption of E-Scooter Sharing in Mid-Sized Swiss Cities

Bachelor Thesis

Submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Digital Business

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1.1 Abstract

Shared electric scooters (e-scooters) have become a visible element of urban mobility across Europe, yet their uptake in mid-sized Swiss cities remains uneven. This thesis investigates the factors that drive residents of three mid-sized Swiss cities — Winterthur, St. Gallen, and Chur — to adopt or reject e-scooter sharing services. Building on the Unified Theory of Acceptance and Use of Technology (UTAUT), a cross-sectional online survey was conducted (N = 214), complemented by two semi-structured interviews with municipal mobility officers. Results indicate that performance expectancy and hedonic motivation are the strongest predictors of intention to use, while perceived safety concerns and unclear parking regulation act as the dominant barriers. Price sensitivity was moderate and strongly age-dependent. The findings suggest that operators and municipalities can raise adoption most effectively by addressing safety perception and infrastructure clarity rather than by lowering per-minute pricing. The thesis contributes a context-specific adoption model for smaller urban centers, where prior research has concentrated on large metropolitan areas.

Keywords: e-scooter sharing, micromobility, technology adoption, UTAUT, Swiss cities

1.2 1. Introduction

1.2.1 1.1 Motivation

Micromobility services promise to close the “last mile” gap between public transport hubs and final destinations. Since 2019, several operators have entered the Swiss market, initially targeting Zurich, Geneva, and Basel. Mid-sized cities followed with a delay, and their experience has been mixed: some pilots were discontinued within a year, while others gradually stabilized. Understanding why residents in these smaller centers adopt – or decline to adopt – shared e-scooters is relevant both for commercial operators deciding where to deploy fleets and for municipalities weighing regulation against mobility benefits.

While a growing body of literature addresses micromobility in large cities, mid-sized cities differ in trip distances, public-transport density, and the availability of alternative modes such as private bicycles. These differences make it unclear whether adoption drivers observed in metropolitan settings transfer to smaller contexts.

1.2.2 1.2 Research Questions

This thesis addresses the following research questions:

- **RQ1:** Which factors most strongly influence the intention to use shared e-scooters among residents of mid-sized Swiss cities?
- **RQ2:** What are the primary barriers that discourage non-users from adopting the service?
- **RQ3:** How do adoption drivers differ across age groups?

1.2.3 1.3 Methodology in Brief

A quantitative online survey grounded in the UTAUT framework was distributed across the three study cities, yielding 214 valid responses. Two expert interviews with municipal mobility officers provided contextual interpretation. Data were analyzed using descriptive statistics and multiple linear regression.

1.3 2. Literature Review

1.3.1 2.1 Micromobility and Urban Transport

Shared micromobility encompasses dockless bicycles, e-bikes, and e-scooters offered on a pay-per-use basis (Abduljabbar et al., 2021). E-scooters are distinguished by low physical effort, short average trip lengths, and strong dependence on smartphone-based access. Studies from European and North American cities report that e-scooter trips frequently substitute for

walking and public transport rather than car trips, complicating their environmental justification (Hollingsworth et al., 2019).

1.3.2 2.2 Technology Acceptance Frameworks

Adoption of digital mobility services is commonly modeled with the Technology Acceptance Model (TAM) and its successor, the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003). UTAUT identifies performance expectancy, effort expectancy, social influence, and facilitating conditions as core determinants of behavioral intention. The extended UTAUT2 adds hedonic motivation, price value, and habit (Venkatesh et al., 2012), which are particularly relevant for discretionary, leisure-adjacent services such as e-scooters.

1.3.3 2.3 Determinants of E-Scooter Adoption

Empirical work has linked e-scooter use to convenience, enjoyment, and time savings (Sanders et al., 2020). Barriers repeatedly identified include safety concerns, weather sensitivity, and regulatory ambiguity regarding riding and parking (Gössling, 2020). Demographic patterns show younger and male riders overrepresented in early adopter groups (Christoforou et al., 2021).

1.3.4 2.4 Research Gap

Most existing studies focus on large metropolitan areas with dense fleets and established regulation. Evidence on mid-sized cities — where fleet sizes are small and public transport is already convenient — is scarce. This thesis addresses that gap by applying a UTAUT2-informed model to three comparable Swiss cities.

1.4 3. Methods

1.4.1 3.1 Research Design

A mixed-methods design was used, with a dominant quantitative survey and a supporting qualitative component. The survey operationalized six UTAUT2 constructs plus a perceived-safety construct added from the barrier literature.

1.4.2 3.2 Sampling and Data Collection

The survey was distributed between March and April 2024 via local community groups, university mailing lists, and printed QR-code flyers placed near transit stops in Winterthur, St. Gallen, and Chur. Of 267 started questionnaires, 214 were complete and valid. Respondents ranged from 16 to 71 years ($M = 34.8$, $SD = 12.6$). Two expert interviews (30–40 minutes each) were conducted with municipal mobility officers in Winterthur and St. Gallen.

1.4.3 3.3 Measures

Constructs were measured on 7-point Likert scales adapted from Venkatesh et al. (2012). Behavioral intention served as the dependent variable. Reliability was acceptable for all constructs (Cronbach's $\alpha > 0.74$).

1.4.4 3.4 Analysis

Descriptive statistics summarized adoption rates and reported barriers. Multiple linear regression estimated the effect of each construct on behavioral intention. Interview transcripts were analyzed thematically.

1.5 4. Results

1.5.1 4.1 Adoption Overview

Of 214 respondents, 41% reported having used a shared e-scooter at least once, while 22% were regular users (at least monthly). Adoption was highest in Winterthur and lowest in Chur.

Table 1. *Self-reported e-scooter usage by city*

City	Respondents	Ever used	Regular users
Winterthur	92	49%	28%
St. Gallen	78	38%	19%
Chur	44	30%	14%
Total	214	41%	22%

1.5.2 4.2 Predictors of Intention

The regression model explained 54% of variance in behavioral intention ($R^2 = 0.54$, adjusted $R^2 = 0.52$). Performance expectancy ($\beta = 0.34$, $p < .001$) and hedonic motivation ($\beta = 0.29$, $p < .001$) were the strongest positive predictors. Perceived safety showed a significant negative association with intention when reversed into a concern score ($\beta = -0.21$, $p < .01$). Price value was significant but weaker ($\beta = 0.12$, $p < .05$).

Figure 1. *Standardized regression coefficients for predictors of behavioral intention*

Performance expectancy	=====	0.34
Hedonic motivation	=====	0.29
Facilitating conditions	=====	0.18
Price value	=====	0.12
Social influence	=====	0.09
Safety concern	-----	-0.21

+-----
-0.3 0 0.1 0.2 0.3 0.4

Note. Bars represent standardized β coefficients from the multiple regression ($N = 214$). Safety concern is reverse-coded and therefore negative.

1.5.3 4.3 Barriers Among Non-Users

Among the 126 respondents who had never used the service, the most cited reasons were safety concerns in traffic (58%), preference for their own bicycle (44%), unclear or inconvenient parking rules (37%), and cost (29%).

1.5.4 4.4 Age Differences

Hedonic motivation was a substantially stronger predictor for the 16–29 group, whereas facilitating conditions (clear rules, reliable app, visible parking) mattered more for respondents over 45. Price value sensitivity decreased with age.

1.6 5. Discussion

The results support a UTAUT2-based reading of e-scooter adoption in mid-sized Swiss cities but with a distinct emphasis. Unlike some metropolitan studies where price value dominates, here performance expectancy and enjoyment led, suggesting that in smaller cities the service competes less on cost and more on convenience and experience against an already strong bicycle culture.

The prominence of safety concern as a barrier (RQ2) aligns with prior findings (Gössling, 2020) and was echoed in both expert interviews, which highlighted unresolved conflicts over sidewalk riding and scattered parking. The age differences (RQ3) imply that a single marketing message is insufficient: younger residents respond to experiential framing, while older residents require reassurance about rules and reliability.

For operators, the practical implication is that infrastructure clarity and safety communication likely yield higher marginal adoption than price promotions. For municipalities, designated parking zones may reduce the most frequently cited friction.

1.6.1 5.1 Limitations

The convenience sample overrepresents younger, digitally connected residents, likely inflating adoption estimates. Self-reported usage is subject to recall bias. The cross-sectional design precludes causal claims. Only three cities were studied, limiting generalizability.

1.7 6. Conclusion

This thesis examined adoption drivers and barriers for shared e-scooters across three mid-sized Swiss cities. Performance expectancy and hedonic motivation emerged as the strongest drivers, while safety concerns and unclear parking regulation were the dominant barriers, with adoption patterns varying by age. The findings suggest that adoption in smaller urban centers hinges more on convenience, experience, and infrastructure clarity than on price. Future research should employ probability sampling and longitudinal designs, and should test whether designated parking interventions measurably increase usage.

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1.9 Appendix A: Survey Instrument (Excerpt)

Selected items (7-point scale, 1 = strongly disagree, 7 = strongly agree):

- PE1: Using a shared e-scooter would help me reach my destination faster.
- HM1: Riding a shared e-scooter would be enjoyable.
- SC1: I would feel unsafe riding a shared e-scooter in traffic.
- FC1: I know where I am allowed to park a shared e-scooter.